LOT-P1 ⍰⍰ 01-BOM

01 — BILL OF MATERIALS & BUYING LIST

Prices: USD planning estimates (qty 1 → qty 100 per unit). **Verify stock/price/footprint before ordering.** Links go to manufacturer/distributor search pages (stable) rather than deep links.

Block	Part	Why	Link	~1 / ~100
MCU+Wi-Fi/BLE	Espressif **ESP32-C3-MINI-1** (13.2×16.6×2.4 mm, pre-certified)	Low power, OTA, TLS, RISC-V	https://www.espressif.com/en/products/modules	3.0 / 2.2
Screen	**Sharp Memory LCD LS013B7DH03** 1.28″ 128×128 mono (≈1 mm)	µW static, readable, thin, pager look	https://www.digikey.com/en/products/result?keywords=LS013B7DH03	14 / 10
Screen alt	1.3″ OLED 128×64 SH1106 (dev)	cheap prototype	https://www.aliexpress.com/w/wholesale-1.3-oled.html	4 / 3
Camera	**Himax HM0360** (VGA, AoN) or **HM01B0** (QVGA) module	Ultra-low-power, tiny	https://www.himax.com.tw/products/cmos-image-sensors/always-on-vision-sensors/	12 / 8
Weather	**Sensirion SHT45** (T/RH)	Accurate humidity	https://www.sensirion.com/products/catalog/SHT45	4 / 2.5
Pressure	**Bosch BMP390**	Barometer/altitude	https://www.bosch-sensortec.com/products/environmental-sensors/pressure-sensors/bmp390/	3 / 1.8
AI gas/air	**Bosch BME688** (BSEC AI, 3×3 mm)	Trainable gas classification	https://www.bosch-sensortec.com/products/environmental-sensors/gas-sensors/bme688/	9 / 6
AI motion	**Bosch BHI260AP** (smart IMU, on-chip ML)	Pick-up / tap / flip, “AI-grade”	https://www.bosch-sensortec.com/products/smart-sensor-systems/bhi260ap/	8 / 5.5
Ambient light	**Vishay VEML7700**	Auto screen contrast	https://www.vishay.com/en/product/84286/	2 / 1.2
Button	Low-profile tact ≤1.5 mm side/edge (e.g. **Alps SKRPACE010**, Omron B3U-1000P)	“Copy”	https://www.digikey.com/en/products/result?keywords=B3U-1000P	0.5 / 0.3
Battery	LiPo **301020–302020** 3.7 V 60–90 mAh, ≤3.0 mm, w/ protection	Fits stack	https://www.digikey.com/en/products/result?keywords=lipo%20302020	6 / 4
Charger IC	**TI BQ25100** (battery charger, 1×1.6 mm)	Safe Li-ion charge	https://www.ti.com/product/BQ25100	1.2 / 0.8
Qi RX	**ST STWLC38** or Renesas P9025AC-class receiver + 20 mm FPC coil + ferrite 0.1 mm	Wireless charge	https://www.st.com/en/power-management/stwlc38.html	5 / 3.5
Coil	RX coil FPC (Würth 760308101 family / custom FPC)		https://www.we-online.com/en/components/products/WCCT	3 / 2
Haptic (opt.)	Piezo haptic driver **DRV2667** + disc, or buzzer	Pager buzz	https://www.ti.com/product/DRV2667	5 / 3.5
LDO/PMIC	TLV75533 3.3 V LDO	Rail	https://www.ti.com/product/TLV75533P	0.4 / 0.2

Block	Part	Why	Link	~1 / ~100
Window	0.5 mm chemically-strengthened glass or sapphire, AR + bezel print	Qi/Wi-Fi/camera/screen	https://www.pcbway.com (custom)	6 / 3
Body	316L back plate + frame, CNC + mirror polish	Req. 3/17	https://www.pcbway.com/cnc-machining-quote.html	90 / 22
PCB	4-layer rigid-flex or 0.8 mm rigid 38×38 mm, ENIG	Main board	https://www.pcbway.com/rigid-flex-pcb.html	(proto 5 pcs ≈ 60) / 4
SMT assembly	PCBWay turnkey		https://www.pcbway.com/pcb-assembly.html	— / 6
Charger (user)	Qi 5 W puck, sized 40 mm, with 316L-matched ring		https://www.digikey.com/en/products/result?keywords=qi%20transmitter%20module	12 / 6

Cost summary (estimate)

	Prototype (5)	100-unit run
Electronics BOM	~$75 ea	~$45 ea
Body + window	~$95 ea	~$25 ea
PCB + SMT + test	~$120 ea	~$14 ea
Charger puck	~$12	~$6
Per unit	**~$300**	**~$90–130**
Batch total (+ NRE tooling/fixtures ~$1.5–2.5k, certs pre-scan ~$1–3k)	~$1.5k	**~$13–18k**

Dev boards for week-1 firmware (buy now): ESP32-C3-DevKitM-1, Sharp memory-LCD breakout (Adafruit 1.3″ 128×128), BME688 + SHT45 + BMP390 breakouts, HM01B0 Arducam module.